NTU EPM & HDAS | 2025 Research Exhibition

Explainable Artificial Intelligence for Personalized Prognosis in Pancreatic Cancer

A Nationwide Study from Taiwan

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Introduction

Background

- ***Pancreatic cancer*** is a highly aggressive malignancy. Due to nonspecific early symptoms and absence of effective screening methods, most patients are diagnosed with *locally advanced* or *metastatic* tumors that are typically *unresectable*.
- In Taiwan, pancreatic cancer was the 7th leading cause of cancer death for males and the **5**th for females in 2022.
  - **2/3** of patients die within 1 year of diagnosis
  - Only **10%** survive beyond 5 years
- Existing artificial intelligence(AI)-based models for pancreatic cancer prognosis:
  - Lack ***interpretability***
  - Overlook feature interactions, non-linear and heterogeneous effects
  - Based on limited or single-institution datasets

Objective

- Develop an accurate and explainable AI prognostic model for pancreatic cancer survival using national Taiwan cancer registry (TCR) data.
- Identify key prognostic factors, their interactions, non-linear relationships, and *patient-specific* survival variability.

Design

TCR Data Flowchart

23,460 Pancreatic cancer patients (**ICD-O-3**: *C25.0 — C25.9*) diagnosed from **2013** to **2021**, with survival outcomes updated until **2022**

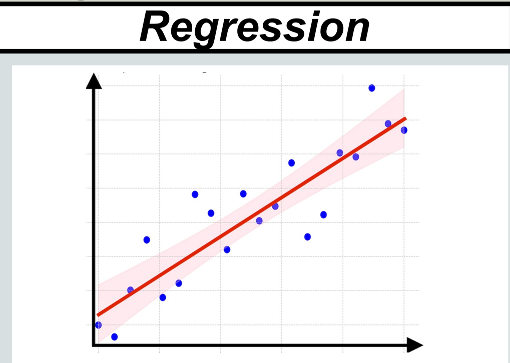
- **13,322** Cases recorded in the **long-form reporting system**, including information about tumor staging, treatments, and site-specific factors
- **10,138** Cases recorded in the **short-form reporting system**

Exclusion Criteria
- Carcinoma in situ, stage 0, or unspecified stage
- Diagnosed age < 20
- Without receiving any curative or palliative treatment
- Duplicate records

8,864 Eligible cases remained for analysis

Prognostic Model Development

Regression



1. Cox PH model with only main effects (***Cox***)
2. Cox PH model with both main effects and all 2-way interaction terms (***Cox-Interact***)

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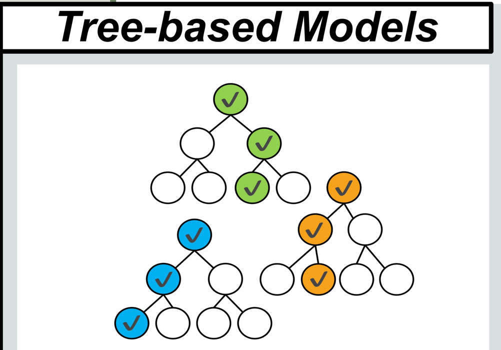
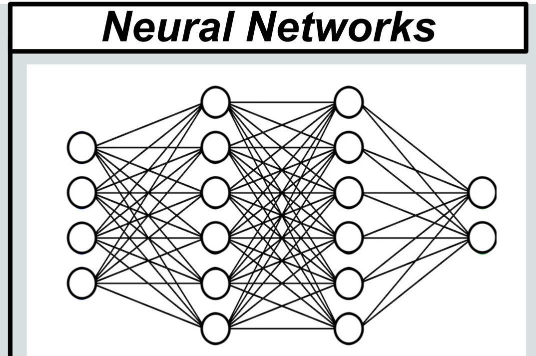
Group LASSO regularization for feature selection

Tree-based Models



1. Oblique random survival forest (***ORSF***)
2. Gradient-boosted trees (***XGBoost***) adapted for censored time-to-event data

Neural Networks



1. ***DeepSurv***
2. ***Cox-Time***
3. ***PC-Hazard***

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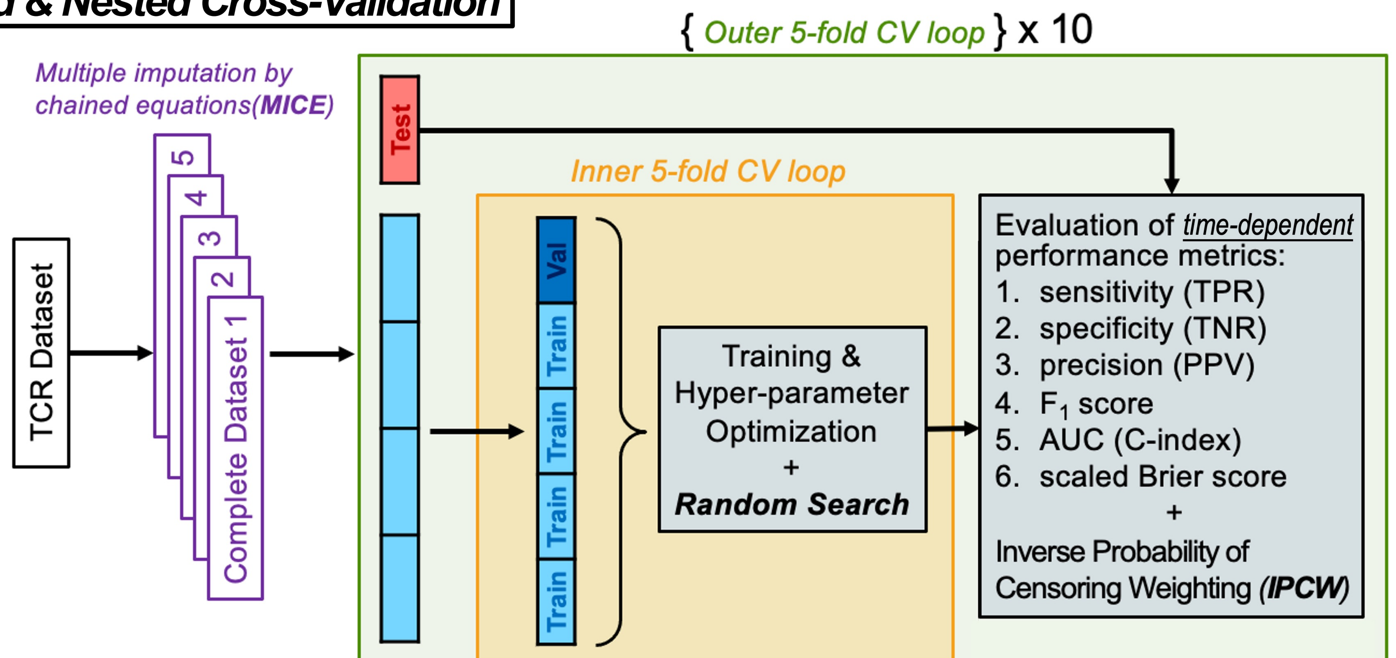
Monotonic network structures defined by 3 tuning parameters:
1. Number of hidden layers
2. Number of nodes in 1st hidden layer
3. A common ratio between the widths of consecutive layers

Model Assessment

Repeated & Nested Cross-Validation



Model Interpretation

***SH**apley **A**dditive ex**P**lanations (SHAP)*

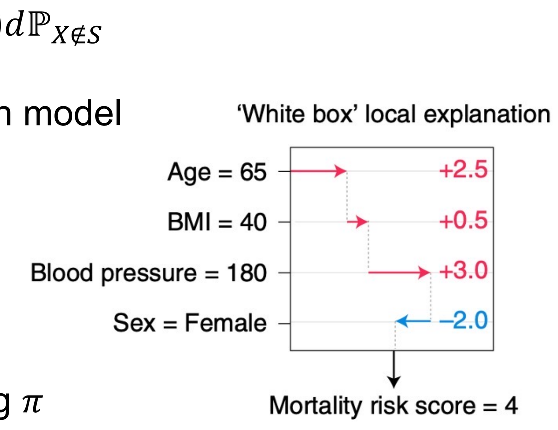
- A ***model-agnostic*** feature attribution method rooted in ***game theory***.
- Quantify a feature's **marginal contribution** by averaging changes in expected model predictions when conditioning on that feature across all possible orderings.
- The **conditional expectation function** of the model's output is defined as

$$g_x(S) = E[\hat{f}(X) \mid X_S = x_S] = \int \hat{f}(x_1, x_2, \dots, x_p) d\mathbb{P}_{X \notin S}$$

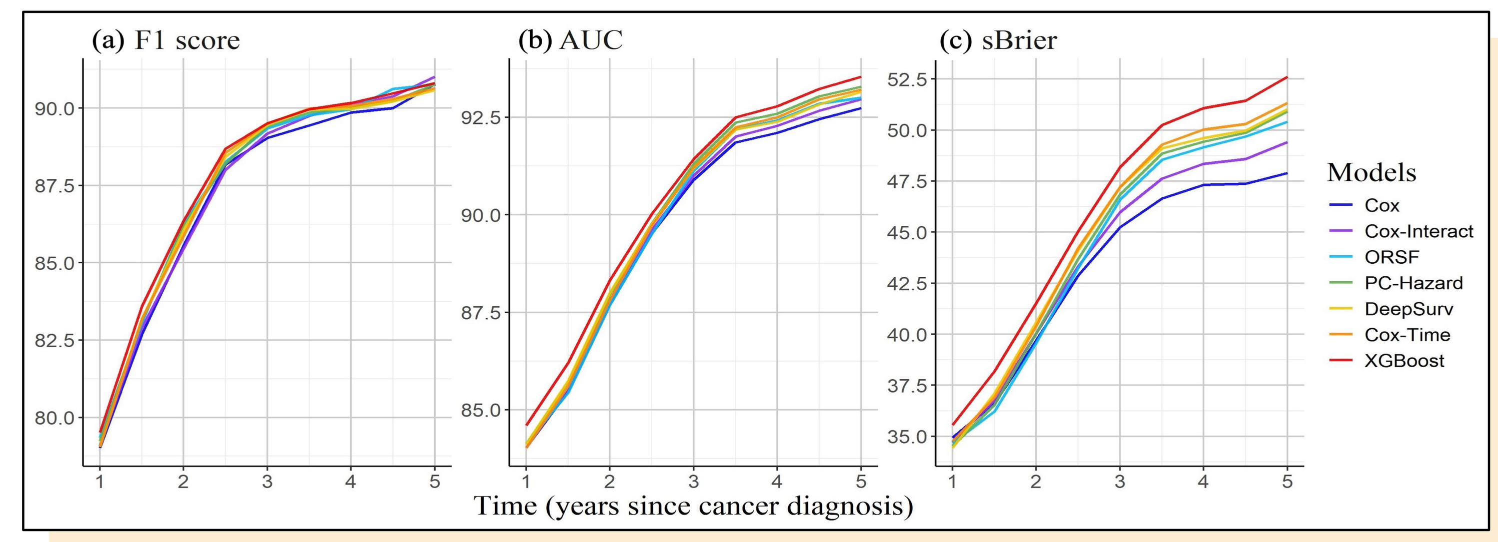
  - S : Set of features we are conditioning on
  - $\hat{f}$: Prediction model
  - x : Model's input vector for the current prediction
- The ***SHAP value*** of feature ***i*** is computed by

$$\phi_i(x) = \frac{1}{|\Pi|} \sum_{\pi \in \Pi} \{g_x(S_i^\pi \cup i) - g_x(S_i^\pi)\}$$

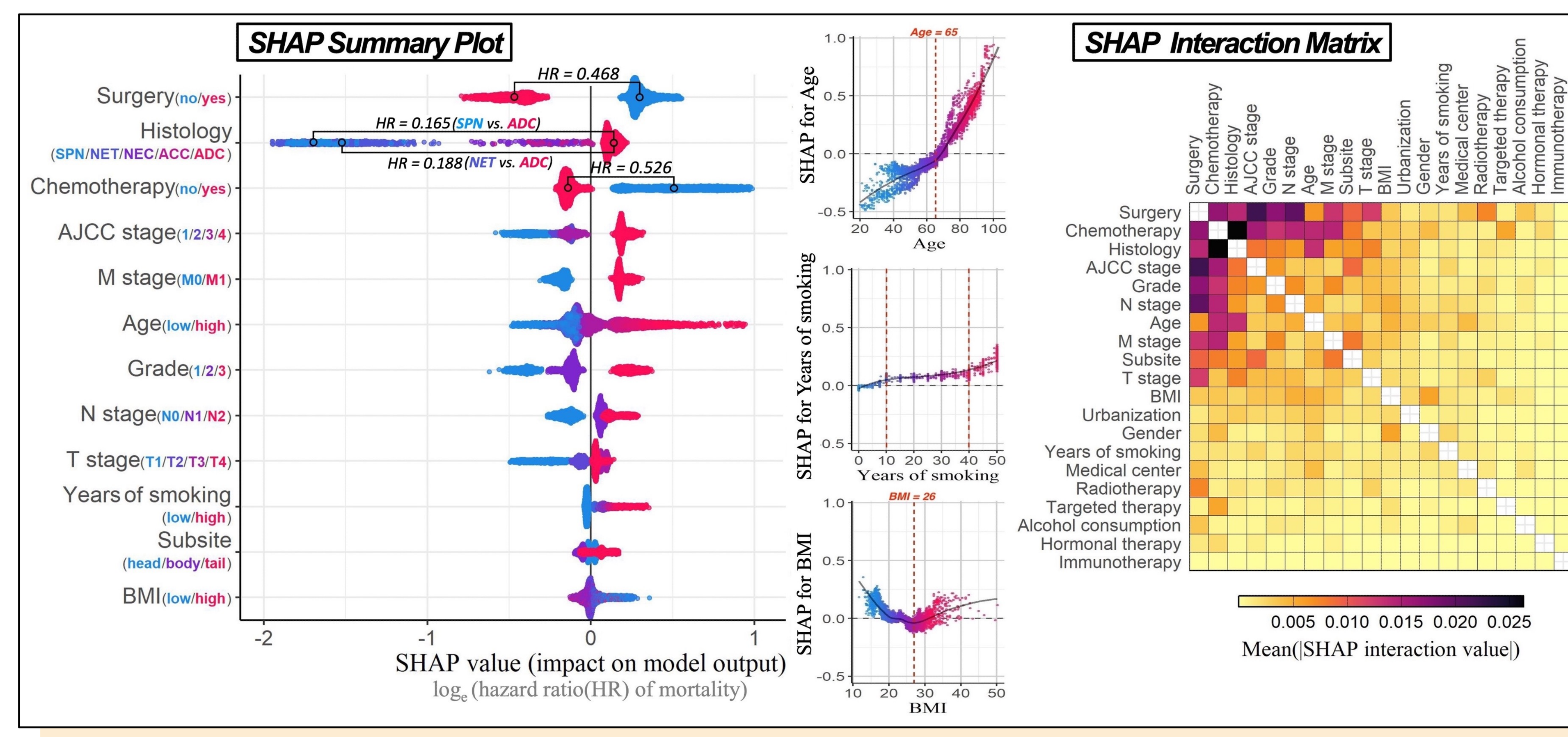
  - Π : Set of all feature orderings
  - S_i^π : Set of all features that come before feature ***i*** in ordering π



Results



- Performance improves across all models as the time since cancer diagnosis increases, plateauing after the 3rd year.
- ***XGBoost*** outperforms traditional Cox models, random survival forest, and advanced deep learning models in predicting pancreatic cancer survival.



- Key determinants for pancreatic cancer survival include ***surgery*** (HR=0.468), ***histological type***, ***chemotherapy*** (HR=0.526), and ***AJCC stage***, and they frequently appear among the most strongly interacting feature pairs.
- **ADC** is linked to a higher mortality risk, while **ACC** and **NEC** have risks slightly below the overall average (HR=0.768, 0.660). In contrast, **NET** and **SPN** have the lowest risks (HR=0.188, 0.165).
- Non-linear relationships for ***age***, ***smoking duration***, and ***BMI*** : mortality increases gradually (0.69% /year) before age 65, rapidly afterward (2.41% /year); smoking risk plateaus after 10–40 years; BMI exhibits a U-shaped risk, lowest at 26.

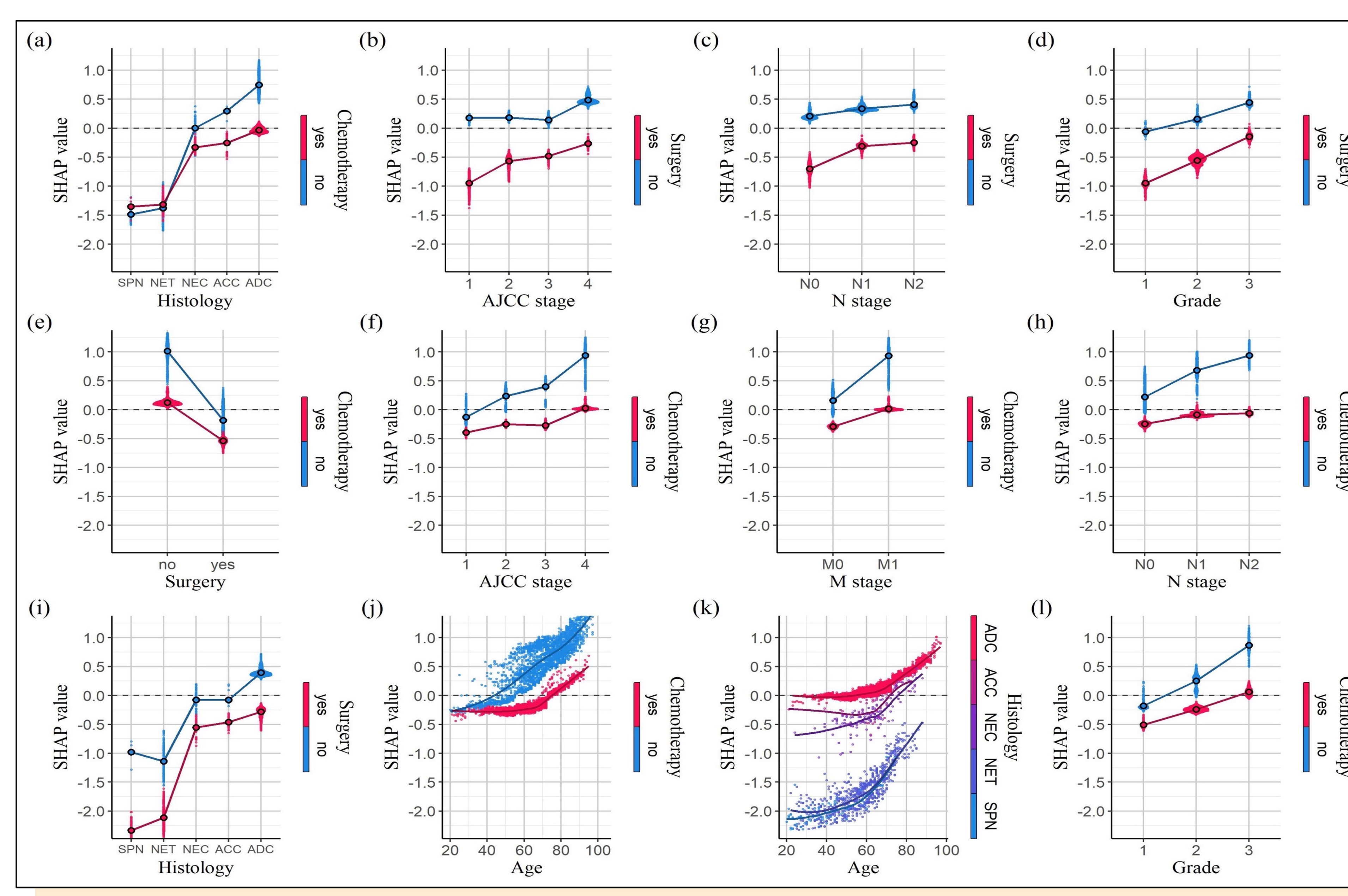


Fig. SHAP dependence plots for the 12 most strongly interacting feature pairs

- Chemotherapy provides the most significant reduction in mortality risk for **ADC** (HR=0.460), but has a more limited effect on **NET**. In contrast, surgery is especially beneficial for both **NET** (HR=0.378) and **ADC** (HR=0.508).
- Surgery most effectively reduces mortality in early-stage cases (*Fig. b-d*), while chemotherapy has a stronger effect in advanced stages (*Fig. f-h, l*).
- Chemotherapy's effectiveness increases with age, notably peaking around 65 years (HR=0.489).
- Mortality risk rises faster with age for **NEC**, **NET**, and **SPN**, especially in older patients.

ADC: adenocarcinoma; NEC: neuroendocrine carcinoma; NET: neuroendocrine tumor; SPN: solid pseudopapillary neoplasm; ACC: acinar cell carcinoma

Discussion

- A log-linear Cox model allows for the explicit inclusion of nonlinear, time-dependent, and interaction terms. However, specifying these terms requires extensive exploratory analysis and statistical testing, increasing the risk of ***model mismatch*** if the assumed structure fails to reflect the true relationships in the data.
- AI-based methods can automatically capture complex nonlinear relationships and interactions in the data. In this study, ***XGBoost*** demonstrated superiority over the 3 deep learning models. Extensive benchmarks have shown that tree-based models consistently outperform neural networks on ***tabular data***.
- Key prognostic factors for pancreatic cancer, such as ***diabetes history***, ***HbA1c***, and tumor markers like ***CEA*** and ***CA19-9***, have only been included in TCR since 2021. Future studies can incorporate these factors into prognostic models to enhance survival predictions and explore their underlying impacts using explainable AI.
- The findings support the use of interpretable AI to personalize treatment strategies, guiding more effective clinical decision-making and informing future development of transparent, scalable prognostic tools in oncology.